The Effect of China's National Forest City Program on Urban Heat Is Not Robustly Identified: Comment on Ma et al. (2026) and Demonstration of the Use of Coding Agents for Replication Studies

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Abstract

Using a staggered adoption difference-in-differences design, Ma et al. (2026) claim that China's National Forest City Program (NFCP) reduces extreme heat days by about three per year in designated cities. However, their data appear to violate parallel trends, they rely on an estimator known to be biased, they provide no evidence for the increase in forest cover that would cause cooling, and their main specification includes plausibly endogenous controls. In the absence of a public replication package, we rely on coding agents to conduct an otherwise impractical replication exercise. Using slightly different data than the original, we replicate their research design using three heterogeneity robust estimators. Point estimates indicate that the NFCP increased extreme heat days and mean summer temperature, although these estimates are imprecise and their 95%CI contains zero. We conclude that the Ma et al. results are not robustly identified.

Keywords: National Forest City Program; urban heat island; Replication and coding agents

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1 Introduction

Ma et al. (2026) use a staggered adoption difference-in-differences design on city-year panel of 243 Chinese prefecture-level cities during 2000–2023 to argue that designation under China's National Forest City Program (NFCP) reduces the annual count of extreme heat days ($\geq$ 35°C) by approximately three per year in designated cities.

There are four reasons to regard this conclusion with skepticism. First, Ma et al. give no evidence that the NFCP increased urban green cover. Second, their main event-study plot shows a decreasing pre-trend, suggesting a violation of the parallel trends assumption on which difference in differences estimation relies. Third, their main estimates use a TWFE regression known to be biased under heterogeneous treatment effects (Goodman-Bacon, 2021). Fourth, their main regression conditions on a long list of covariates, per capita GDP among them. Because per capita GDP is plausibly correlated with the propensity to apply for NFCP designation and temperature trends (e.g. from industrial or commercial development) treatment effect estimates may be confounded.

We repeat the Ma et al. (2026) evaluation of the NFCP using three heterogeneity robust estimators. First, the Borusyak et al. (BJS) imputation estimator with covariate adjustment (Borusyak et al., 2024). Second, the Callaway–Sant'Anna estimator (CS) (Callaway and Sant'Anna, 2021), which does not permit covariate adjustment. Third, its doubly robust variant (DR) (Sant'Anna and Zhao, 2020), with covariate adjustment.[1]

We find no evidence that the NFCP increased forest cover. If anything, satellite-measured vegetation near city centers declines after NFCP designation. We also find that, using a satellite based temperature measure, our point estimates for the effect of the NFCP on summer mean temperature are slightly positive, indistinguishable from zero, and precise enough to allow us to reject a decrease in summer mean temperature of more than about 0.2°C. Repeating these estimations using weather station, rather than remotely sensed temperature

[1] BJS without covariate adjustment does not satisfy the parallel pre-trends condition and we do not report it.

data, and using the sample of 99 cities for which weather station data is publicly available, we find a statistically insignificant increase of about 0.1°C in mean summer temperature. If we use the same extreme heat days outcome as Ma et al. (2026), instead of satellite or instrument based measures of summer temperature, we find that NFCP increased annual hot days by about 1.6, also indistinguishable from zero.

2 Data and Methods

Data

We draw our city-level panel from the United Nations World Urbanization Prospects (WUP 2018) dataset (United Nations, Department of Economic and Social Affairs, Population Division, 2018). These data describe cities with populations of at least 300,000 and WUP centroids anchor all satellite data extractions. Treatment is a binary indicator equal to one for city i in all years at or after its first NFCP designation, constructed from annual lists published by the National Forestry and Grassland Administration. The panel covers 275 prefecture-level cities observed from 2000 to 2025 (7,150 city-years); 149 hold NFCP designation across 17 cohorts (2004–2022) and 126 remain never-treated. We extract all outcomes at buffer radii of 5, 10, and 20 km from the city centroid. Main estimates use the 20 km buffer; 5 km results are reported in Table 1 and in Appendix A1.

China has approximately 337 prefecture-level administrative units. Our sample of 275 is smaller for two reasons. First, the WUP includes only agglomerations exceeding its 300,000-resident threshold; smaller prefecture-level units fall below it. Second, we exclude county-level cities that appear in the WUP because the NFCP programme targets prefecture-level units exclusively.

To measure temperature, we use MODIS 8-day composite LST at 1 km resolution (Wan et al., 2021) (downloaded, NASA EOSDIS, June 2026.). For each city-year we average June–August pixels within each buffer around the city center, excluding fill values. Note

two deviations from Ma et al. (2026). First, the Ma et al. (2026) outcome is calculated on the basis of prefecture boundaries rather than disks centered on the city center. Second, the Ma et al. (2026) outcome is a weather station derived measure of days per year warmer than 35°C rather than a satellite based summer average temperature. We verify that our conclusions are robust to calculating outcomes using prefecture boundaries and a subsample of weather station data to match Ma et al. (2026) more closely. We cannot exactly match the Ma et al. (2026) sample because the full sample of weather stations is not publicly available.

We consider three satellite based measures of urban vegetation and tree cover. First, MODIS reports the Normalized Difference Vegetation Index (NDVI) monthly composites at 1 km resolution (Didan, 2021)(downloaded, NASA EOSDIS, June 2026). From these data we compute mean July NDVI for each city-year. Second, the MODIS Vegetation Continuous Fields product reports annual tree canopy cover at 250 m resolution (DiMiceli et al., 2021) (downloaded, NASA EOSDIS, June 2026). From these data we compute mean canopy fraction for each city-year. Finally, the CLCD reports annual Landsat-based land cover classifications at 30 m resolution (Yang and Huang, 2021) (Zenodo record 18180184, 1985–2025, downloaded June 2026). From these data we compute the forest pixel share for each city-year. With a 30 m resolution, the CLCD detects land cover changes not visible in the lower resolution MODIS data.

The panel is largely complete. Among the 7,150 city-years, MODIS LST at 5 km has 286 missing observations (4.0%), concentrated in persistently cloudy cities; LST at 20 km, NDVI, and tree cover have fewer than 30 missing observations combined. CLCD is complete for all 7,150 city-years.

We supplement our satellite based city-year analysis with a second city-year analysis in which outcomes are derived from weather station records. Ma et al. (2026) use station data from the Chinese Meteorological Administration (CMA), which is not publicly available. We use publicly available GHCN-Daily records (Menne et al., 2012) instead. We retain stations within 50 km of a city centroid with at least 30 valid daily maximum temperature readings per

year; city-years for which no station meets this threshold are excluded from the estimation sample. Where a city has multiple qualifying stations, all outcomes are averaged across stations to yield one observation per city-year. The station sample covers 52 treated cities (16 cohorts) and 47 never-treated cities. Temperature outcomes are summer mean daily maximum temperature (Tmax, June–August) and the count of days with Tmax ≥ 35°C. Vegetation outcomes are MODIS NDVI and tree cover extracted from the pixel closest to each station, and CLCD forest cover as the forest pixel share within a 100 m radius buffer around each station; where multiple stations exist within a city, these per-station values are averaged to one city-year observation.[2]

Methods

Ma et al. (2026) implement a staggered adoption differences in differences design. Their main estimates use OLS-TWFE with a treated $\times$ post indicator. With heterogenous treatment effects, this estimator may be biased (Goodman-Bacon, 2021). The literature proposes a number of heterogeneity robust estimators for staggered adoption designs. We implement three of them and find that the Ma et al. (2026) result disappears.

The main regression in Ma et al. (2026) conditions on a set of time-varying city-level controls, including per capita GDP. Per capita GDP is plausibly correlated with both the propensity to apply for NFCP designation and with temperature trends independent of the program. Therefore, the effects of this control may be confounded with the effects of the treatment.

In a difference-in-differences design, the identifying assumption is parallel trends: absent designation, treated and control cities would have followed the same outcome path. Researchers include covariates to restore conditional parallel trends when the unconditional version fails (or to improve precision when it holds). We find that BJS without covariate adjustment shows pre-designation LST coefficients that diverge from zero. Conditioning on

[2]Temperature outcomes use all 99 cities, satellite outcomes (LST, NDVI, tree cover, and CLCD) at station locations use 82–84 cities because of missing satellite data.

pre-determined city characteristics (latitude, longitude, and log pre-treatment population) absorbs this imbalance.

The BJS imputation estimator (Borusyak et al., 2024) fits TWFE on untreated observations (never- and not-yet-treated units), imputes counterfactual outcomes for treated units, and identifies treatment effects as observed minus imputed. BJS with controls augments the first stage TWFE with covariate $\times$ time interactions for city latitude, longitude, and log pre-treatment population, allowing cities to follow different time trends. These covariates are pre-determined and time-invariant. Because they are time-invariant, entering them in levels would be collinear with city fixed effects; we therefore interact each with five-year period-bin dummies. The Callaway–Sant'Anna estimator (Callaway and Sant'Anna, 2021) compares each treatment cohort to never-treated cities at each event-time horizon without covariate adjustment. Finally, the doubly robust estimator (Sant'Anna and Zhao, 2020) is a variant of CS that allows for covariates. DR conditions on the same pre-determined variables (latitude, longitude, and log pre-treatment population). For each estimator we aggregate to an average post-period ATT as the simple mean of event-study coefficients at $t = 0, 1, \ldots, 10$, with the mean of period-level clustered standard errors as a conservative SE approximation.

The period-level standard errors for BJS with controls specifications (columns 1–2 of table 1) are computed in a way that treats the imputed counterfactuals as known rather than estimated, and therefore does not implement the efficient influence-function variance derived in Borusyak et al. (2024) (Section 4). Implementing the influence-function based variance calculation for the BJS estimator with controls is computationally demanding and is not part of the Stata package that implements this estimator. Because all BJS-with-controls temperature estimates are far from any conventional significance threshold, this probably does not affect any substantive conclusion. Standard errors for DR (column 3 of table 1) and CS (column 4 of table 1) are from Stata's `csdid`, which implements the influence-function variance of Sant'Anna and Zhao (2020) and Callaway and Sant'Anna (2021) respectively.

The code that organized the data and performed the estimations in this comment was

written by a Claude agent. To validate the data processing pipeline, we first confirmed that all outcome values lay in plausible ranges. Second, for each raw data layer, we constructed a synthetic layer for which the final city-year level outcomes were known and verified that these values obtained. To validate the estimation routines, we first conducted all estimations twice, once using python estimation libraries, and once using stata estimation routines. We next gave a separate Claude agent, with no knowledge of the project, the final data set and a draft of this comment, and confirmed that the agent could reproduce results from these sources alone.

MODIS land surface temperature (MOD11A2), NDVI (MOD13A3), and tree cover (MOD44B) are publicly available from NASA EOSDIS. The China Land Cover Dataset (CLCD) is publicly available from Zenodo (record 18180184; Yang and Huang 2021). GHCN-Daily station temperature records are publicly available from NOAA (Menne et al., 2012). NFCP designation lists are publicly available from the National Forestry and Grassland Administration of China. A replication package is available; see Turner (2026).

3 Results

Table 1 reports average post-period ATTs. The Ma et al. (2026) temperature result does not appear. BJS with controls gives +0.032°C (95% CI [−0.18, +0.24]) at 20 km and +0.016°C (95% CI [−0.22, +0.25]) at 5 km, both indistinguishable from zero. DR gives +0.010°C (95% CI [−0.17, +0.19]) and CS gives +0.051°C (95% CI [−0.14, +0.24]). For reference, the mean absolute year-to-year change in summer mean LST across cities is 1.0°C; for hot days (≥ 35°C) across stations it is 6.2 days. In sum, no estimate of NFCP on temperature is statistically distinguishable from zero, point estimates are positive rather than negative, and magnitudes are small relative to normal annual variation.

The remaining two columns consider variants of our specification that more nearly resemble Ma et al. (2026). Column 5 replicates the results of column 4 using outcomes calculated

Table 1: Average post-period treatment effects ($t = 0$ to $t = 10$). Each cell is the simple mean of event-study coefficients at $t = 0, 1, \ldots, 10$. Standard errors in parentheses are the mean of period-level clustered standard errors. 95% confidence intervals [ATT $\pm$ 1.96×SE] are shown in square brackets for LST and hot days. Columns: (1) BJS with controls at 20 km; (2) BJS with controls at 5 km; (3) DR with controls at 20 km; (4) CS at 20 km; (5) CS at prefecture-level geography; (6) BJS with controls, outcomes from weather station records (city-year, averaged across stations).

	(1) BJS w/ctrl, 20 km	(2) BJS w/ctrl, 5 km	(3) DR 20 km	(4) CS 20 km	(5) CS pref	(6) BJS w/ctrl, stn
LST (°C)§	+0.032	+0.016	+0.010	+0.051	+0.055	+0.114
	(0.106)	(0.121)	(0.093)	(0.095)	(0.087)	(0.184)
	[−0.18, +0.24]	[−0.22, +0.25]	[−0.17, +0.19]	[−0.14, +0.24]	[−0.12, +0.23]	[−0.25, +0.47]
NDVI (index)	+0.002	−0.001	−0.000	+0.000	−0.001	−0.019
	(0.002)	(0.003)	(0.002)	(0.002)	(0.002)	(0.018)
Tree cover (%)	+0.095	+0.047	−0.116	−0.196	−0.146	−1.060
	(0.287)	(0.380)	(0.248)	(0.265)	(0.253)	(0.998)
CLCD forest (pp)	−0.194	−0.585*	−0.096	−0.263	−0.188	−2.41
	(0.301)	(0.319)	(0.201)	(0.196)	(0.178)	(2.07)
Hot days (count)	—	—	—	—	—	+1.602
						(2.062)
						[−2.44, +5.64]
Covariates	Y	Y	Y	N	N	Y

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors clustered at the city level. § In column (6), the temperature row reports station-measured summer mean daily maximum temperature (Tmax, °C), not MODIS LST.

on the basis of prefecture boundaries rather than 5 or 20km disks. CS gives +0.055°C (95% CI [−0.12, +0.23]), consistent with other results. Column 6 replicates results in column 1 for the 99 cities with qualifying weather station data. The analysis remains at the city-year level; where multiple stations fall within the 50 km buffer, outcomes are averaged across stations. Temperature outcomes use station-measured Tmax and hot days rather than MODIS LST; vegetation outcomes use the nearest satellite pixel to each station rather than averaging over circular buffer zones. Row 1 reports station summer mean Tmax and row 5 reports hot days ($\geq$ 35°C), the same outcome as Ma et al. (2026). As for all other columns, point estimates are small and indistinguishable from zero.

Rows 2, 3, and 4 of table 1 report treatment effects of NFCP designation on measures of forest cover. Except for the outcome variables, the same specifications are used as for temperature outcomes. MODIS NDVI, tree cover, and CLCD forest cover show no post-

designation increase at any buffer radius. In fact, CLCD forest share declines near city centers. BJS with controls at 5 km gives -0.585 pp ($SE = 0.319$, $p < 0.10$); CS and DR estimates are statistically null but uniformly negative. These results indicate that the NFCP probably fails to increase forest cover and may decrease it.

Figure 1 illustrates these findings. The four panels show BJS with controls event studies using outcomes averaged over 20 km radius disks centered on the city center. The top-left panel (summer LST) shows point estimates that scatter around zero from the start of the pre-period through $t = 10$, with confidence intervals that contain zero throughout. The remaining three panels show no obvious post-designation change in MODIS NDVI, MODIS tree cover fraction, or Landsat-derived CLCD forest cover. All of the outcomes are noisy and not obviously affected by NFCP designation. A possible exception is the CLCD based forest cover plot in the bottom right. Here, pre-trends look parallel and there is a weak decline in tree cover post treatment. In sum, these plots demonstrate that the findings in Ma et al. (2026) are not robust.

Appendix Figures A1, A2, A3, and A4 present event study figures like those in figure 1, but corresponding to columns 2, 3, 5, and 6 of Table 1. Each tells more or less the same story. Pre-period and post period estimates are imprecisely estimated and scattered around zero. The CLCD event studies in the bottom right of figures A1 and A2 are partial exceptions. These figures show that treatment with NFCP designation may have decreased forest cover. That A1 also suggests a decreasing pretrend invites the conjecture that cities with decreasing central forest cover selected into the program and continued on this trend after designation. None of the event studies supports the Ma et al. (2026) claim that NFCP designation caused a reduction in urban temperature.

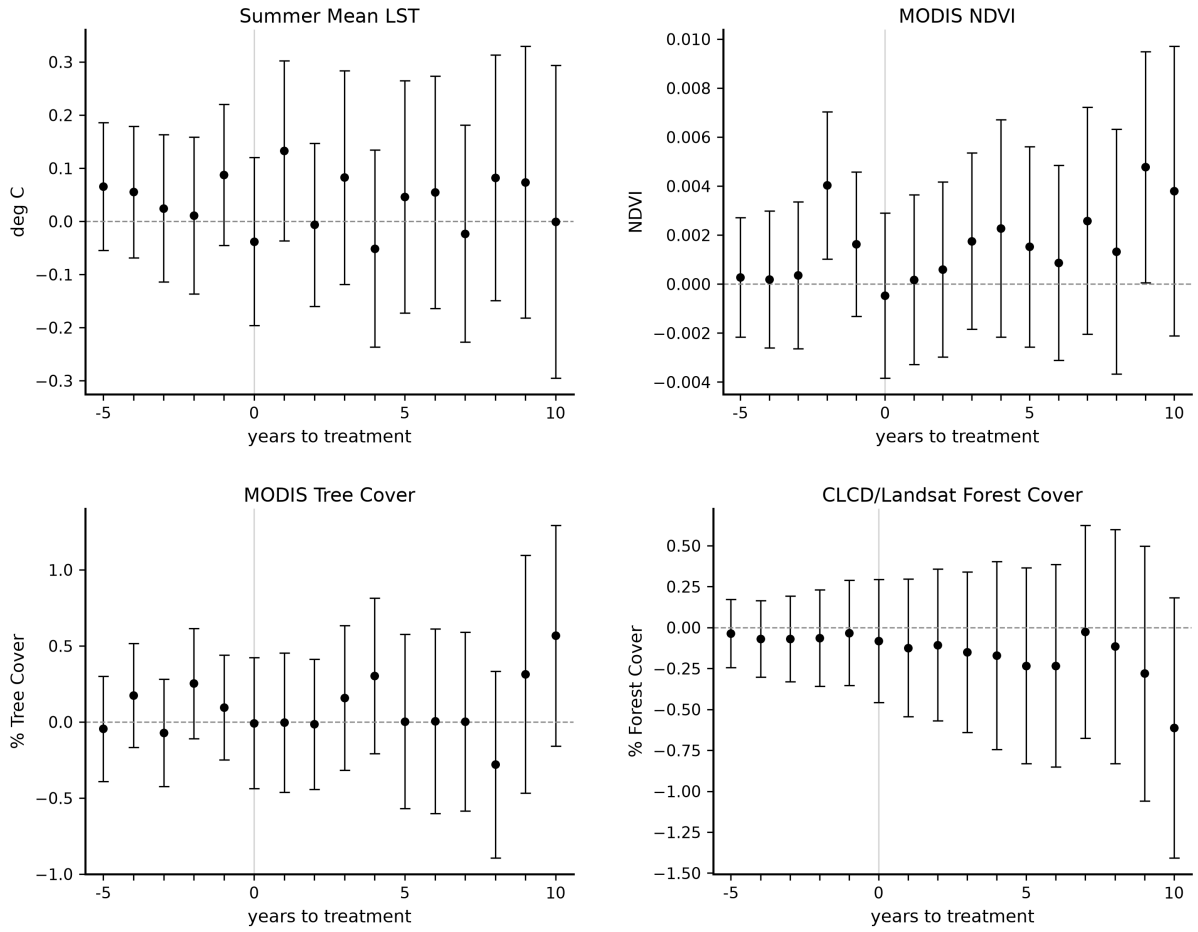


Figure 1: BJS imputation with covariate adjustment (BJS with controls) event studies for outcomes averaged over pixels within 20 km of the city centroid, corresponding to column (1) of Table 1. The horizontal axis shows years to treatment ($t = 0$ is the designation year, $t = -1$ is the reference period). Dots are point estimates; vertical whiskers show the 95% confidence interval clustered at the city level. *LST (°C)*: top left. *NDVI (index)*: top right. *Tree cover (%)*: bottom left. *CLCD forest (pp)*: bottom right.

4 Conclusion

The temperature effect of NFCP designation is not robustly identified under any of the heterogeneity robust estimators that we consider. Across all specifications the estimates of the effect of the NFCP on temperature are slightly positive, statistically insignificant, and allow us to rule out anything except a small cooling effect. The Ma et al. (2026) temperature result does not appear in our analyses.

The vegetation results add an independent argument against the proposed mechanism.

MODIS NDVI, MODIS tree cover, and CLCD forest cover show no post-designation increase at any buffer radius. CLCD forest share declines marginally at 5 km. Without an increase in forest cover at any radius, there is no plausible mechanism by which the NFCP could reduce urban heat.

We conclude that the main result in Ma et al. (2026) is not robustly identified.

References

Kirill Borusyak, Xavier Jaravel, and Jann Spiess. Revisiting event-study designs: Robust and efficient estimation. *Review of Economic Studies*, 91(6):3253–3285, 2024. doi: 10.1093/restud/rdae007.

Brantly Callaway and Pedro H. C. Sant'Anna. Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230, 2021. doi: 10.1016/j.jeconom.2020.12.001.

Kamel Didan. MOD13A3 MODIS/Terra Vegetation Indices Monthly L3 Global 1km SIN Grid V061, 2021.

Compton DiMiceli, Mark Carroll, Robert Sohlberg, Do Hyung Kim, Maggi Kelly, and John R. G. Townshend. MOD44B MODIS/Terra Vegetation Continuous Fields Yearly L3 Global 250m SIN Grid V061, 2021.

Andrew Goodman-Bacon. Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277, 2021. doi: 10.1016/j.jeconom.2021.03.014.

Yun Ma, Yan Geng, and Shuai Zhong. Urban forest development and extreme heat mitigation: The climate adaptation effects of China's National Forest City Policy. *Forests*, 17 (1):79, 2026. doi: 10.3390/f17010079.

Matthew J. Menne, Imke Durre, Russell S. Vose, Byron E. Gleason, and Tamara G.

Houston. An overview of the Global Historical Climatology Network-Daily database. *Journal of Atmospheric and Oceanic Technology*, 29(7):897–910, 2012. doi: 10.1175/JTECH-D-11-00103.1.

Pedro H. C. Sant'Anna and Jun Zhao. Doubly robust difference-in-differences estimators. *Journal of Econometrics*, 219(1):101–122, 2020. doi: 10.1016/j.jeconom.2020.06.003.

Matthew Turner. Replication package for "The Effect of China's National Forest City Program on Urban Heat Is Not Robustly Identified", 2026. URL `https://github.com/m-turner/Ma_etal_Forests2026_replication`.

United Nations, Department of Economic and Social Affairs, Population Division. World urbanization prospects: The 2018 revision, file 22 – population of urban agglomerations with 300,000 inhabitants or more. Technical report, United Nations, New York, 2018.

Zhengming Wan, Simon Hook, and Glynn Hulley. MOD11A2 MODIS/Terra Land Surface Temperature/Emissivity 8-Day L3 Global 1km SIN Grid V061, 2021.

Jie Yang and Xin Huang. The 30 m annual land cover dataset and its dynamics in China from 1990 to 2019. *Earth System Science Data*, 13(8):3907–3925, 2021. doi: 10.5194/essd-13-3907-2021.

Appendix

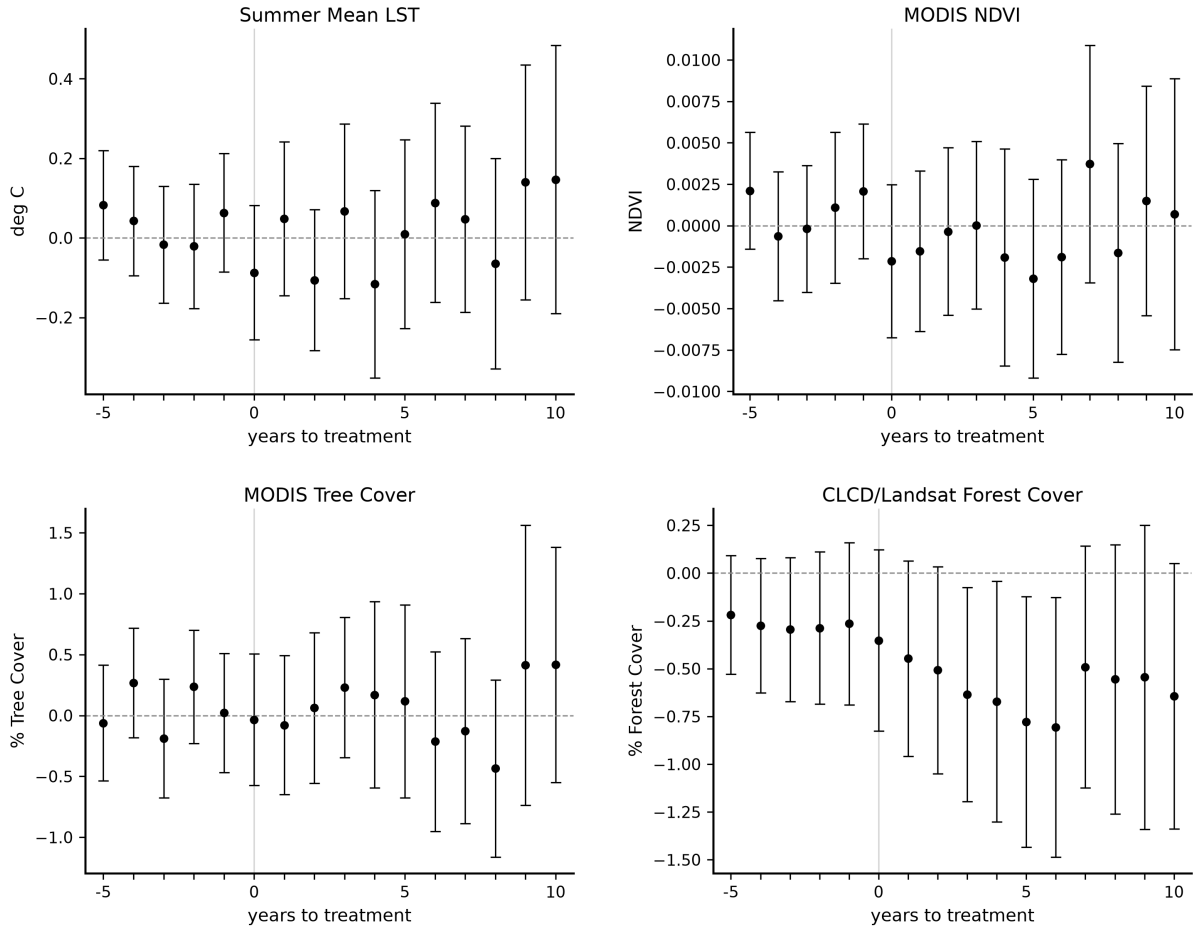


Figure A1: BJS imputation with covariate adjustment (BJS with controls) event studies at the 5 km buffer, corresponding to column (2) of Table 1. The horizontal axis shows years to treatment ($t = 0$ is the designation year). Dots are point estimates; vertical whiskers show the 95% confidence interval clustered at the city level. *LST (°C)*: top left. *NDVI (index)*: top right. *Tree cover (%)*: bottom left. *CLCD forest (pp)*: bottom right.

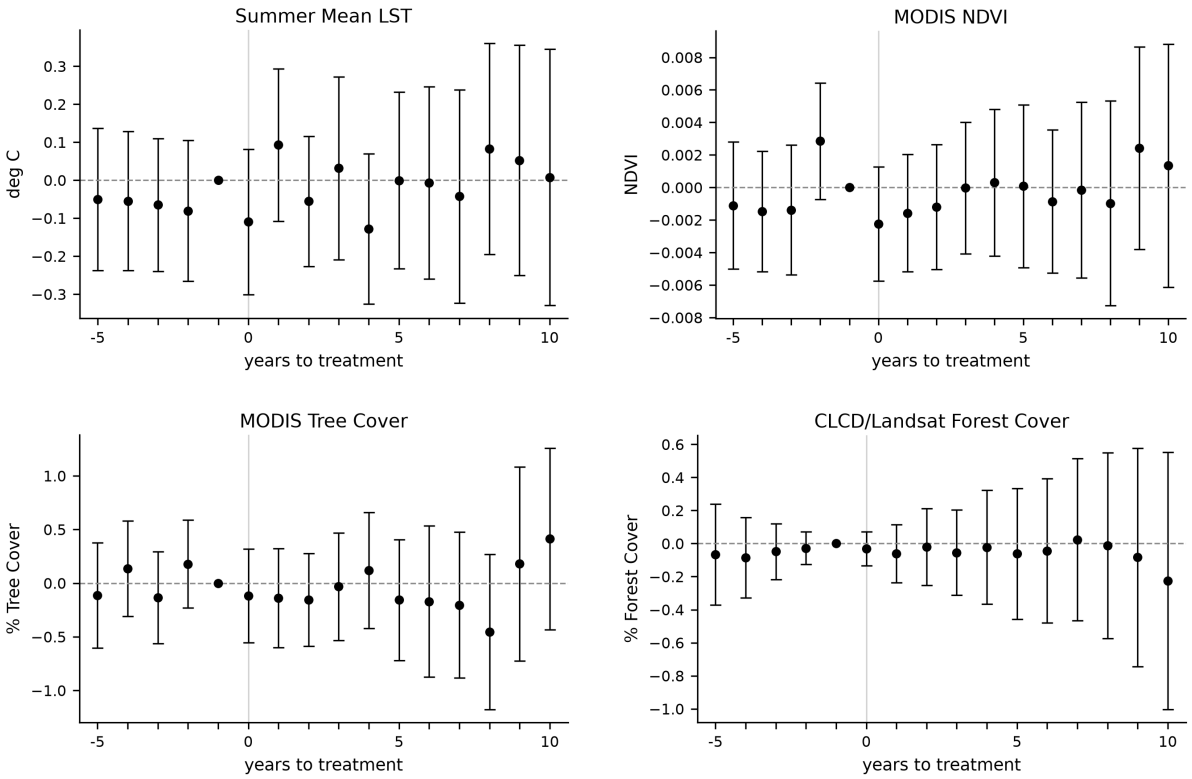


Figure A2: Doubly robust Callaway–Sant'Anna (DR) event studies at 20 km conditioning on pre-determined city characteristics (latitude, longitude, and log pre-treatment population), corresponding to column (3) of Table 1. The horizontal axis shows years to treatment ($t = 0$ is the designation year, $t = -1$ is the reference period). Dots are point estimates; vertical whiskers show the 95% confidence interval from Stata's `csdid` (Callaway and Sant'Anna, 2021; Sant'Anna and Zhao, 2020). *LST (°C)*: top left. *NDVI (index)*: top right. *Tree cover (%)*: bottom left. *CLCD forest (pp)*: bottom right.

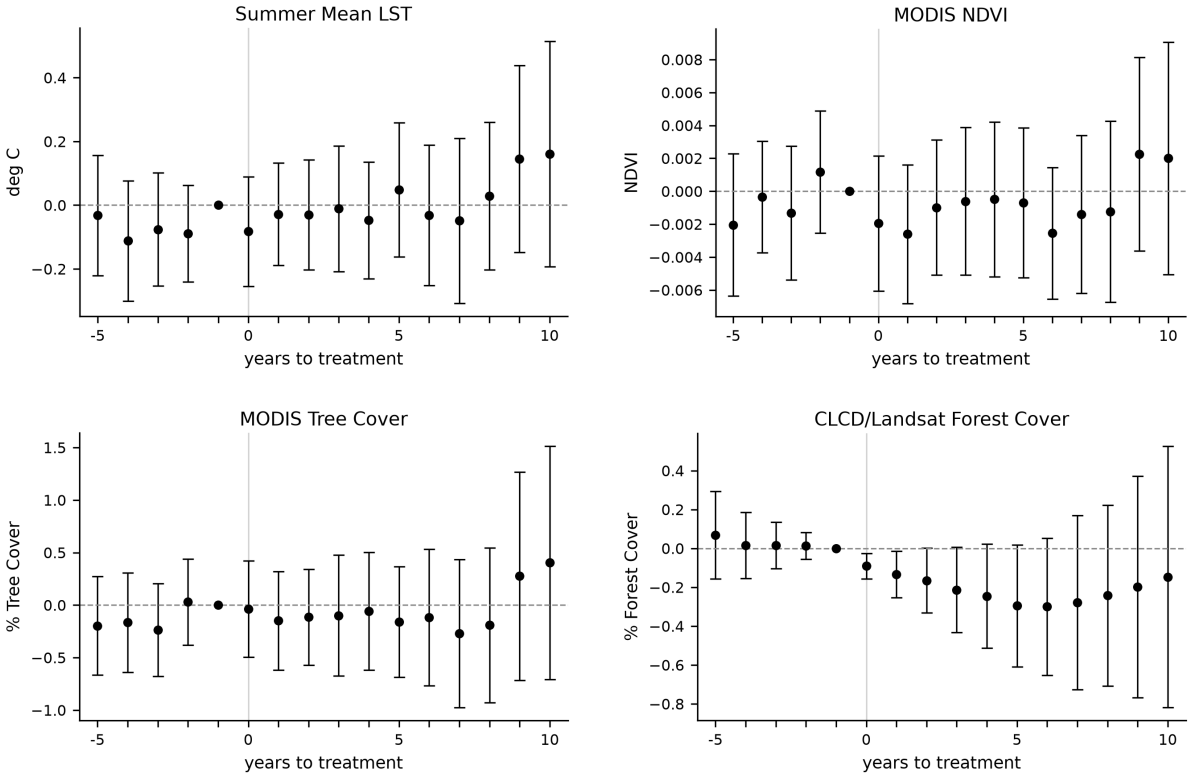


Figure A3: Callaway–Sant'Anna (CS, no covariate adjustment) event studies at the prefecture-level geography, corresponding to column (5) of Table 1. The horizontal axis shows years to treatment ($t = 0$ is the designation year, $t = -1$ is the reference period). Dots are point estimates; vertical whiskers show the 95% confidence interval from Stata's `csdid` (Callaway and Sant'Anna, 2021). *LST (°C)*: top left. *NDVI (index)*: top right. *Tree cover (%)*: bottom left. *CLCD forest (pp)*: bottom right.

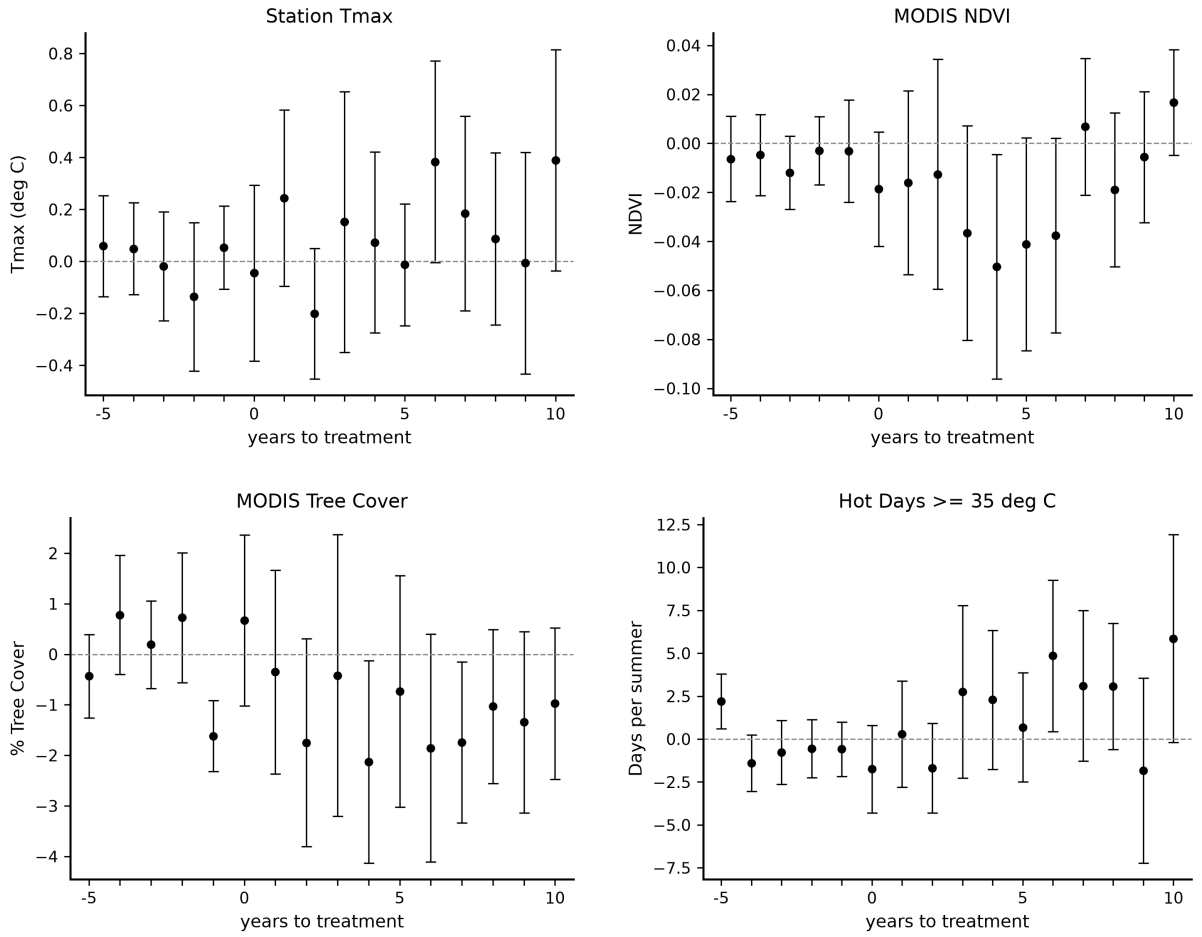


Figure A4: Station-level BJS imputation with covariate adjustment (BJS with controls) event studies, corresponding to column (6) of Table 1. The horizontal axis shows years to treatment ($t = 0$ is the designation year, $t = -1$ is the reference period). Dots are point estimates; vertical whiskers show the 95% confidence interval via sandwich estimator. Top left: summer mean daily maximum temperature. Top right: MODIS NDVI at nearest pixel to each station. Bottom left: MODIS tree cover at nearest pixel Bottom right: Annual days with Tmax ≥ 35°C.